■ Guardian AI – Financial Fraud & Scam Advisor Chatbot

■ Full Development Process (Simple but Complete)

■ 1. Data Collection

Goal: Collect all types of data related to online frauds and scams.
- Gather text data (fraud messages, fake job offers, phishing links)
- Collect image and document data (fake IDs, invoices, screenshots)
- Use public fraud datasets (Scamwatch, PhishTank, Spamhaus, FTC)
- Label each data item (Phishing, Fake Job, Crypto Scam)
- Clean, organize, and store data securely.

■ 2. Model Development

Goal: Build AI models that can detect, score, and explain scams.
Models to Develop:
1. Fraud Classification Model – Detects the type of scam.
2. Trust Scoring Model – Gives a 0–100 fraud risk score.
3. Global Scam Check Model – Verifies with global scam databases.
4. Document & Image Authenticity Model – Checks if documents/images are fake.
5. Deepfake Detection Model – Detects edited or AI-generated media.
6. Explanation Model – Explains why something is flagged as fraud.

■ 3. Model Training & Testing

Goal: Make sure models work correctly and perform well in real time.
- Split data into training, testing, and validation sets.
- Train models on labeled data.
- Measure accuracy, F1-score, and recall.
- Optimize models for fast response (under 1.5 sec).
- Test using real-world scam samples.

■ 4. API Integration

Goal: Connect all AI models to the backend through APIs.
- Use FastAPI or Flask to create API routes.
- Example routes: /analyze_text, /trust_score, /verify_document, /detect_deepfake, /explain_fraud
- Send results in JSON format.
- Connect APIs to the chatbot backend.

■ 5. Chatbot Development

Goal: Build an AI chatbot that talks with users and gives instant fraud analysis.
- Design a friendly chat interface.
- Use NLP (Dialogflow, Rasa, or LLM) for natural understanding.
- Allow text, link, or image inputs.
- Chatbot connects to backend APIs and returns results with fraud type, score, and advice.

Frontend Tools: Flutter (Mobile) / React (Web).

■ 6. Backend Development

Goal: Manage all chatbot requests and AI model responses.
- Build backend with FastAPI or Node.js.
- Connect to models, databases, and authentication.
- Use MongoDB or PostgreSQL for storage.
- Secure backend with JWT, SSL, and logging.

■ 7. Cloud Setup & Deployment

Goal: Host everything online for real-time use.
- Host AI models on AWS SageMaker or Google Vertex AI.
- Host backend on AWS Lambda, EC2, or Cloud Run.
- Store media in AWS S3.
- Use CloudWatch or Grafana for monitoring.

■ 8. Testing & Quality Check

Goal: Ensure the system works smoothly and safely.

Types of Testing:
- Unit Testing
- Integration Testing
- Performance Testing
- Security Testing

■ 9. Launch & Continuous Learning

Goal: Release chatbot to public and keep improving models.
- Deploy mobile and web apps.
- Collect user feedback.
- Add new scam data and retrain models.
- Regularly update and redeploy.

■ End Result

You get a 24×7 AI Fraud Prevention Chatbot that can: - Analyze messages, links, and images instantly. - Give fraud type, trust score, and human-readable explanation. - Help users avoid scams and report them easily.